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Roll No 23i-5066

Assignment Activity Ratios

Submitted to Sir Akhtar

Section "4A"

103/120

Date	DD	MM	YY

"ACTIVITY RATIOS"

Q NO:01

Receivable turnover = $\frac{\text{Credit sales or Net Sales}}{\text{Average gross recievables}}$

For 2002 = $\frac{17000}{1700}$ = 10 times

For 2003 = $\frac{14580}{1620}$ = 9 times

For 2004 = $\frac{9600}{1600}$ = 6 times

For 2005 = $\frac{9000}{1800}$ = 5 times

Receivable turnover in days = $\frac{\text{Average gross receivables}}{\text{Net Sales}} \times 365$

For 2002 = $\frac{1700}{17000} \times 365$ = 36.5 days

For 2003 = $\frac{1620}{14580} \times 365$ = 40.6 days

For 2004 = $\frac{1600}{9600} \times 365$ = 60.8 days

For 2005 = $\frac{1800}{9000} \times 365$ = 73 days

1) Performance of collection department from year 2002 to 2005-

* Receivable turnover has decreased from 10 to 5, meaning the company is collecting payments at a slower rate.
* Collection period has increased from 36.5 to 73 days, showing weaker efficiency in recovering recievables.

2) Change in collection trends:- The collection period has nearly doubled, indicating delayed payments from customers.

Potential causes: Lax credit policies, customer delayed payments, over-reliance on credit sales.

QNo:02

1) Average Account receivable = $\frac{\text{Beginning receivables + Ending receivables}}{2}$

$$= \frac{160000 + 180,000}{2} = 170,000$$

2) Account receivable turnover = $\frac{\text{Net sales}}{\text{Average accounts receivables}}$

10

$$= \frac{31,50,000}{170,000} = 18.53$$

3) Account receivable turnover in days = $\frac{\text{Avg accounts receivables}}{\text{Net sales}} \times 365$

$$= \frac{170,000}{31,50,000} \times 365 = 19.698$$

4) Days receivable turnover = $\frac{\text{Ending gross receivables}}{\text{Net Sales}} \times 365$

$$= \frac{180,000}{31,50,000} \times 365 = 20.857$$

) Average Inventory = $\frac{\text{Beginning inventory + ending inventory}}{2}$

$$= \frac{480,000 + 390,000}{2} = 435,000$$

Inventory turnover = $\frac{\text{Cost of goods sold}}{\text{Avg Inventory}}$

$$= \frac{22,50,000}{435,000} = 5.17$$

Inventory turnover in days = $\frac{\text{Avg Inventory}}{\text{Cost of goods sold}} \times 365$

$$= \frac{435,000}{22,50,000} \times 365 = 70.567$$

6) Number of Days sales in inventory = $\frac{\text{Ending Inventory}}{\text{Cost of goods sold}} \times 365$

$$= \frac{480,000}{2250,000} \times 365 = 77.867$$

7) Operating cycle = Account receivable turnover in days + ~~Inventory~~ turnover in days.

$$= 19.698 + 70.567 = 90.265 \text{ days.}$$

Q No. 03:

1) Average Inventory = $\frac{\text{Beginning} + \text{Ending}}{2}$

For 1987 $= \frac{173,999 + 195,512}{2} = 184,756$

For 1986 $= \frac{167,286 + 173,999}{2} = 170,643$

2) Inventory turnover = $\frac{\text{Cost of goods sold}}{\text{Avg Inventory}}$ =

For 1987 $= \frac{786,523}{184,756} = 4.26$ times

For 1986 $= \frac{700,263}{170,643} = 4.10$ times

3) Inventory turnover in days = $\frac{\text{Avg inventory}}{\text{cost of goods sold}} \times 365$

For 1987 $= \frac{184,756}{786,523} \times 365 = 85.7$ days

For 1986 $= \frac{170,643}{700,263} \times 365 = 89.$ days

4) Number of Days sales in inventory = $\frac{\text{Ending inventory}}{\text{Cost of good sold}} \times 365$

For 1987 $= \frac{195,512}{786,523} \times 365 = 90.73$ days

For 1986 $= \frac{173,999}{700,263} \times 365 = 90.69$ days

5) Working capital turnover = $\dfrac{\text{Net sales}}{\text{Avg Working capital}}$

For 1987 $= \text{avg} = \dfrac{241{,}952 + 266586}{2} = 254{,}269 = \dfrac{1086{,}944}{254{,}269}$

For 1986 $= \text{avg} = \dfrac{217{,}848 + 241{,}952}{2} = 229{,}900$

$= \dfrac{988{,}417}{229900} = 4.30$

6) Working capital turnover in days = $\dfrac{\text{Avg working capital}}{\text{Net Sales}} \times 365$

For 1987 $= \dfrac{254{,}26{,}9}{1086{,}944} \times 365 = 85.3$ days

For 1986 $= \dfrac{229900}{988\,417} \times 365 = 84.9$ days

7) Number of Days sales invested in working capital = $\dfrac{\text{Ending working capital}}{\text{Net Sales}} \times 365$

For 1987 $= \dfrac{266{,}586}{1086{,}944} \times 365 = 89.52$ days

For 1986 $= \dfrac{241{,}952}{988{,}417} \times 365 = 89.34$ days

Comments :-

- Inventory turnover improved : increased from 4.10 to 4.26, indicating better stock movement.
- Faster inventory sales : inventory turnover in days decreased from 89 to 85.7, showing quicker sales.
- Stable working capital : Minimal change 4.30 to 4.28 reflecting consistent sales efficiency.
- Efficient working capital use : Days sales in working capital remained nearly the same 85.3 vs 84.9 days, showing steady operational efficiency.

Q04:-

1) (a) Cost of goods sold:

COGS = Sales − Gross profit
$= 125{,}000 - 25{,}000 = 100{,}000$

b) Closing Stock = Opening + Purchases − COGS
$= 20{,}000 + 110{,}000 - 100{,}000 = 30{,}000$

10

2) Period of credit given to debtors
(Accounts receivable turnover in days)

Accounts Receivable turnover $= \frac{\text{Sales}}{\text{Debtors}} = \frac{125{,}000}{25{,}000} = 5$

Accounts receivable turnover in days $= \frac{25{,}000}{125{,}000} \times 365 = 73$ days

3) Accounts Payable turnover $= \frac{\text{Purchases on credit}}{\text{Creditors}} = \frac{110{,}000}{22{,}000} = 5$

4) The Rate of Stock turnover (Inventory turnover) $= \frac{\text{COGS}}{\text{Average Inventory}}$

Avg inventory $= \frac{20{,}000 + 30{,}000}{2} = 25{,}000$

Inventory turnover $= \frac{100{,}000}{25{,}000} = 4$

5) Inventory turnover in days

$= \frac{25000}{100{,}000} \times 365 = 91.25$

Q No:05

Total asset turnover $= \frac{\text{Net Sales}}{\text{Avg Total assets}}$

10

Year 1 $= \frac{10{,}50{,}000}{2{,}30{,}000} = 4.57$ times

Year 2 $= \frac{10{,}00{,}000}{200{,}000} = 5$ times

Q No:06

10

1) Inventory turnover and impact of Poor management:-
Inventory turnover measures how often stock is sold and replaced.
High turnover: efficient management, lower storage costs.
Low turnover: overstocking tied up capital, risk of obsolete stock.

⇒ Poor Inventory leads to :- Higher holding costs, cash flow p... reduced profitability.

2) Inventory turnover $= \frac{COGS}{\text{Average stock}}$ & Inventory turnover in days :- $= \frac{\text{Average}}{COGS}$

$2002 = \frac{15000}{1000} = 15$ times	$2002 = \frac{1000}{15000} \times 365 = 24.3$ days
$2003 = \frac{9600}{1000} = 9.6$ times	$2003 = \frac{1000}{9600} \times 365 = 38$ days
$2004 = \frac{14580}{1620} = 9$ times	$2004 = \frac{1620}{14880} \times 365 = 40.6$ days
$2005 = \frac{17{,}000}{1740} = 9.8$ times	$2005 = \frac{1740}{17000} \times 365 = 37.2$ days

QNO:07:-

03

1) Accounts receivable turnover $= \frac{\text{Sales}}{\text{Avg. Accounts Receivables}}$

$$2010 = \frac{1{,}500{,}000}{115{,}000} = 13.04$$

$$2011 = \frac{1{,}400{,}000}{107{,}500} = 13.02$$

2) Accounts receivable turnover in days $= \frac{\text{Avg. Accounts receivables}}{\text{Sales}} \times 365$

$$2010 = \frac{115{,}000}{1{,}500{,}000} \times 365 = 28.8 \text{ days}$$

$$2011 = \frac{107{,}500}{1{,}400{,}000} \times 365 = \cancel{28.4 \text{ days}}$$

3) Inventory turnover $= \frac{COGS}{\text{Avg. Inventory}}$

$$2010 = \frac{1{,}020{,}000}{240{,}000} = \cancel{5.1}\ 4.25$$

$$2011 = \frac{1{,}120{,}000}{225{,}000} = \cancel{4.48}\ 4.98$$

Inventory turnover in days = $\frac{\text{Avg. Inventory}}{\text{COGS}} \times 365$

$$2010 = \frac{240{,}000}{1{,}020{,}000} \times 365 = 71.6 \text{ days}$$

86

73

$$2011 = \frac{285{,}000}{1{,}120{,}000} \times 365 = 81.5 \text{ days}$$

5) Operating cycle = accounts receivable turnover in days + Inventory turnover in days.

114

$$2010 = 26.8 + 71.6 = 98.4 \text{ days}$$

101

$$2011 = 27.4 + 81.5 = 108.9 \text{ days}$$

QNO:08:.

10/2

1) Debtors turnover $= \frac{\text{Sales}}{\text{Debtors}}$

$$= \frac{3{,}100{,}000}{770{,}000} = 4.03$$

2) Debtors collection Period (in Days) $= \frac{\text{Debtors}}{\text{Sales}} \times 365$

$$= \frac{770{,}000}{3{,}100{,}000} \times 365 = 90.6 \text{ days}$$

3) Cost of goods sold

COGS = Sales − Gross Profit

$$= 3100{,}000 - 1{,}725{,}000 = 1{,}375{,}000$$

4) Stock turnover $= \frac{\text{COGS}}{\text{Stocks}}$

$$= \frac{1{,}375{,}000}{310{,}000} = 4.43$$

5) Stock turnover in days $= \frac{\text{Stocks}}{\text{COGS}} \times 365 = \frac{310{,}000}{1{,}375{,}000} \times 365 = 82.4 \text{ days}$

6) Operating cycle = Debtors collection Period + Stock turnover in days

$$= 90.6 + 82.4 = 173 \text{ days}$$

Trend Analysis.

* A high debtor collection period (90.6 days) suggets delayed pay. which may affect liquidity.
* The inventory turnover (4.43 times) is moderate, but holding inventory for 82.4days could indicate slow-moving stock.
* The overall operating cycle of 173 days is quite long, meaning cash conversion is slow.
* To improve efficiency, the company should speed up collections and optimize inventory management.

QNO:09

a) Gross accounts receivable at the beginning for the year 2011 & 2010 - ~~2011.~~

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Gross = Net + Allowance

2011 = 240,360 + 12 300 = 252,660

2010 = 230,180 + 7180 = 237,360

b) Gross accounts receivables at the end =

2011 = 220,385 + 11,180 = 231,565

2010 = 240,360 + 12,300 = 252,660

c) Average gross receivables for the Year

Average gross =

$$2011 = \frac{252,660 + 231,565}{2} = 242,112.5$$

$$2010 = \frac{237,360 + 252,660}{2} = 245,010$$

d) Accounts receivable turnover (Using average gross receivables)

$$\text{AR turnover} = \frac{\text{Net sales}}{\text{Average gross receivables}}$$

$$2011 = \frac{1180,178}{242,112.5} = 4.88 \text{ times}$$

$$2010 = \frac{2,200,000}{245,010} = 8.98 \text{ times}$$

Accounts receivable turnover in days $= \dfrac{365}{\text{AR turnover}}$

$$2011 = \frac{365}{4.88} = 74.8 \text{ days}$$

or

$$2011 = \frac{242112.5}{1180,178} \times 365 = 74.8 \text{ days}$$

$$2010 = \frac{245,010}{2,200,000} \times 365 = 40.6 \text{ days}$$

f) Days sales in Receivables

$$2011 = \frac{231,565}{1180,178} \times 365 = 71.6 \text{ days}$$

$$2010 = \frac{252,660}{2,200,000} \times 365 = 41.91 \text{ days}$$

g) AR turnover using beginning gross receivables $= \dfrac{\text{Net sales}}{\text{Beginning gross receivables}}$

$$2011 = \frac{1180178}{252,660} = 4.67 \text{ times}$$

$$2010 = \frac{2,200,000}{237,360} = 9.27 \text{ times}$$

In 2010, Hawk company collected receivables in about 41 days (= 9 turns/year). By 2011, the collection period increased to roughly 75 days (= 5 turns/year), indicating a decline in liquidy and slower cash collections.

Ques:

Q NO: 10:-

a) Gross receivables = Net receivables + Allowance

2007 = 640360 + 12300 = 652660

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2006 = 530180 + 7180 = 537360

b) Ending:

2007 =

620,385 + 11780 = 631565

2006 =

540360 + 12300 = 552660

c) Avg gross receivables = $\frac{\text{Beginning} + \text{Ending}}{2}$

2007 = $\frac{652660 + 631565}{2} = 642113$

2006 = $\frac{537360 + 552660}{2} = 545010$

d) Acc receivable turnover = $\frac{\text{Net Sales}}{\text{Avg receivables}}$

2007 = $\frac{2180178}{642113} = 3.40$

2006 = $\frac{2300000}{545010} = 4.22$

e) Turnover in days = $\frac{\text{Avg receivables}}{\text{Net Sales}} \times 365$

2007 = $\frac{642113}{2180178} \times 365 = 107.35$ days

2006 = $\frac{545010}{2300000} \times 365 = 86.49$ days

f) Days sales = $\frac{\text{ending}}{\text{Net sales}} \times 365$

2007 = $\frac{631565}{2180178} \times 365 = 105.7$ days

2006 = $\frac{552660}{2300000} \times 365$

= 87.7 days

Acc receivables turnover using beginning = $\frac{\text{Net sales}}{\text{Beginning}}$

$$2007 = \frac{2180178}{652660} = 3.34 \text{ times}$$

$$2006 = \frac{2300000}{537360} = 4.28 \text{ times}$$

* Receivables decline from 4.28 to 3.34. This means the company is collecting receivables more slowly in 2007 compared to 2006.
* lower turnover rates suggests that more capital is held up in accounts receivables.
* Company is taking longer to convert its receivables into cash

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QNO:011:-

1 - Accounts receivable turnover = $\frac{\text{Net Sales}}{\text{Avg acc receivables}}$

2) Avg acc receivables = $\frac{\text{Beginning} + \text{Ending}}{2}$

$$= \frac{280,000 + 255000}{2} = 267500$$

$$\text{turnover} = \frac{3050600}{267500} = 11.41 \text{ times}$$

3) 2 - turnover in days = $\frac{\text{Avg acc receivables}}{\text{Net Sales}} \times 365$

$$= \frac{267500}{3050600} \times 365 = 32 \text{ days}$$

3 - Days sales = $\frac{\text{ending}}{\text{Net Sales}} \times 365$

$$= \frac{255000}{3050600} \times 365 = 30.5 \text{ days}$$

4 - Inventory turnover = $\frac{\text{COGS}}{\text{Avg Inventory}}$

Avg Inventory = $\frac{565000 + 523000}{2} = 544000$

turnover $= \frac{2185100}{544000} = 4.02$ times

5) Inventory turnover in days $= \frac{\text{Avg inventory}}{\text{COGS}} \times 365$

$= \frac{544000}{2185100} \times 365$

$= 90.8$ days

6) Days sales $= \frac{\text{Ending}}{\text{COGS}} \times 365$

$= \frac{523000}{2185100} \times 365$

$= 87.4$ days

7) Operating cycle = acc receivable in days + inventory in days

$= 32 + 90.8$

$= 122.8$ days

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106/100

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Assignment #3.

Submitted to Sir Akhtar

Section BS-BA-B

Date	03	03	2025

able Turnover =

$$\frac{\text{Net Sales}}{\text{Average Gross Profit Receivables}}$$

2002	2003	2004	2005
10 times	9 times	6 times	5 times.

Receivable Turnover in days =

$$\frac{\text{Average Gross Account Receivable}}{\text{Net Sales}} \times 365.$$

10

2002	2003	2004	2005
37 days.	41 days	61 days	73 days

Question 2:-

(1) $\frac{\text{Opening + Closing}}{2}$

$$\frac{180,000 + 160,000}{2}$$

$= 170,000$

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(2) $\frac{\text{Net Sales}}{\text{Avg Gross Receivables}}$

$$\frac{3150,000}{170,000}$$

$= 19$ times

(3) $\frac{\text{Average Gross Receivables}}{\text{Net Sales}} \times 365$

$$\frac{170,000}{31,50,000} \times 365$$

$= 20$ days.

(4) $\frac{\text{Ending Gross Receivables}}{\text{Net Sales}} \times 365$

$\frac{180,000}{31,50,000} \times 365$

= 21 days.

(5) $\frac{\text{opening} + \text{closing}}{2}$

$\frac{390,000 + 480,000}{2}$

= 435,000.

(6) $\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$

$\frac{22,50000}{435,000}$

5.172 times.

(7) $\frac{\text{Average Inventory}}{\text{Cost of Goods Sold}} \times 365.$

$\frac{435,000}{22,50,000} \times 365$

71 days.

(8) $\frac{\text{Ending Inventory}}{\text{Cost of Goods Sold}} \times 365$

$\frac{480,000}{22,50,000} \times 365$

= 78 days.

(9) Acc receivable turnover in days + Inventory turnover in days.

20+71

= 91 days.

$\frac{\text{+ Closing}}{2}$

$186\text{:-} \frac{167{,}286 + 173{,}999}{2}$

$= 170{,}643$

$1987 = \frac{173{,}999 + 195{,}512}{2}$

$= 184756$

(2) $\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$

$1986\text{:-} \frac{700{,}263}{170{,}643}$

$= 4 \text{ times}$

$1987 = \frac{786{,}523}{184{,}756}$

$= 4{\cdot}25 \text{ times}$

(3) $\frac{\text{Average Inventory}}{\text{Cost of Goods Sold}} \times 365$

$1986 = \frac{170{,}643}{700{,}263} \times 365$

$= 89 \text{ days}$

$1987 = \frac{184{,}756}{786{,}523} \times 365$

$= 86 \text{ days}$

(4) $\frac{\text{Ending Inventory}}{\text{Cost of Goods Sold}} \times 365$

$1986 = \frac{173{,}999}{700{,}263} \times 365$

$= 91 \text{ days}$

$1987 = \frac{195{,}512}{786{,}523} \times 365$

$= 91 \text{ days}$

(5) $\frac{\text{Net Sales}}{\text{Average working capital}}$

$$1986 = \frac{988{,}417}{\frac{(217{,}848 + 241{,}952)}{2}} = \frac{988{,}417}{229900}$$

$= 4.3$ times

$$1987 = \frac{1086944}{254269}$$

$= 4.3$ times

(6) $\frac{\text{Average Working Capital}}{\text{Net Sales}} \times 365$

$$1986 = \frac{229900}{988{,}417} \times 365$$

$= 85$ days

$$1987 = \frac{254269}{1086944} \times 365$$

85.3 days

8

(7) $\frac{\text{Ending Working Capital}}{\text{Net Sales}} \times 365$

$$1986 = \frac{241{,}952}{988{,}417} \times 365$$

$= 89$ days

$$1987 = \frac{266{,}586}{1086944} \times 365$$

$= 90$ days

Sales - Gross Profit.

125,000 - 25,000

= 100,000

(b) (Opening Stock + Purchases) - Cost of Goods Sold.

130,000 - 100,000

Closing Stock = 30,000

2. $\frac{\text{Average Gross Receivables}}{\text{Net Sales}} \times 365$

$\frac{25,000}{125,000} \times 365$

= 73 days

3. $\frac{\text{Credit Purchase}}{\text{Accounts Payable}}$

$\frac{110,000}{22,000}$

= 5 times.

4. $\frac{\text{Accounts Payable}}{\text{Credit Purchase}} \times 365$

$\frac{22,000}{110,000} \times 365$

= 73 days.

5. $\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$

Avg Inventory = $\frac{20,000 + 30,000}{2}$

= 25,000

$\frac{100,000}{25,000}$ = 4 times

COGS = Sales - Gross Profit.

= 125,000 - 25,000

= 100,000

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(6) $\frac{\text{Average Inventory}}{\text{CoGS}} \times 365$.

$\frac{25,000}{100,000} \times 365$

= 91.25 days.

Question 5:-

$\frac{\text{Net Sales}}{\text{Average Asset Turnover}}$

Y1 = $\frac{10,50,000}{230,000}$ = 5 times

Y2 = $\frac{10,00,000}{200,000}$ = 5 times.

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Question 6:-

1. Inventory turnover measures how efficiently a business sells and replaces its stock within a given period. A higher inventory turnover indicates strong sales and efficient inventory management, while a lower turnover suggests slow-moving stock and poor management.
Bad inventory management can increase storage and holding costs of business. It can cause cash flow problems.

2. Inventory Turnover

$\frac{\text{Cost of Goods Sold}}{\text{Average Stock}}$

2005	9.77 times
2004	9.0 times
2003	9.6 times
2002	15.0 times

Inventory Turnover in days = $\frac{\text{Average Stock}}{\text{CoGS}} \times 365$

2005	37 days
2004	40 days
2003	38 days
2002	24 days

1

The highest inventory turnover was in 2002 which was 15 days, showing better stock management. By 2005, it declined to 9.77 times, with inventory days rising from 24 to 37 days, indicating slower sales and weaker inventory management, which may increase costs and reduce profitability.

...nt Receivable

2010	2011

$$\frac{15,00\,000}{1101,000} = 13.64 \text{ times} \qquad \frac{14\,00,000}{105,000} = 13.33 \text{ times}$$

2) $\frac{\text{Account Receivable}}{\text{Sales}} \times 365$

2010

$\frac{110,000}{1500000} \times 365$

26.76 days

2011

$\frac{105,000}{1400000} \times 365$

$= 27.375$ days.

3) Inventory Turnover =

$\frac{\text{CoGS}}{\text{Inventory}}$

2010

$\frac{1020,000}{200,000}$

$= 5.1$ times

2011

$\frac{11,20,000}{250,000}$

$= 4.48$ times

4) $\frac{\text{Inventory}}{\text{COGS}} \times 365$

2010

$\frac{200,000}{1020,000} \times 365$

$= 71.568$ days

2011

$\frac{250,000}{1120,000} \times 365$

$= 81.473$ days.

5) A/R turnover in days + Inventory turnover in days.

2010

27+72

= 99 days

2011

= 27+81

= 108 days

TB

Question 8:-

(i) $\frac{\text{Sales}}{\text{Account Receivables}}$

$\frac{31,00,000}{770,000}$ = 4 Times

(ii) $\frac{\text{Account Receivables}}{\text{Net Sales}} \times 365$

$\frac{770,000}{31,00,000} \times 365$ = 91 days

(iii) Sales - Gross Profit.

31,00,000 - 17,25,000

= 1375,000

(iv) $\frac{\text{COGS}}{\text{Stocks}}$

$\frac{1375,000}{310,000}$

= 4 times.

(v) $\frac{\text{Stocks}}{\text{COGS}} \times 365$

$\frac{310,000}{1375,000} \times 365$

= 82 days

(vi) A/R turnover in days + Inventory turnover in days

91 + 82

= 173 days

...ceivables = Net Receivables + Allowance for losses.

...0= 237360

...011= 240,360+12,300= 252660

2010 = 252660

2011 = 231565

(c) Avg Gross Receivables

2010 = 245010

2011 = 242113

(d) A/R Turnover =

2010 = $\frac{2200,000}{245010}$ = 9 Times

2011 = $\frac{1180178}{242113}$

= 5 Times

(e) 2010

$\frac{245010}{2200,000} \times 365$

= 41 days.

2011

$\frac{242113}{1180178} \times 365$

= 75 days

(f) 2010 = $\frac{252660}{2200,000} \times 365$

= 42 days.

2011 = $\frac{231565}{1180,178} \times 365$

= 72 days.

(g) 252660

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Question 10:-

(a) Account Receivable at the Beginning:-

2006 = 530,180 + 7180 = 537360

2007 = 640,360 + 12,300 = 652660

(b) Account Receivable at the end of year:-

2006 = 540,360 + 12,300 = 552660

2007 = 620,385 + 11,180 = 631565

(c) Average Gross Receivables:-

2006 = 537360 + 552660 = 1090020

2007 = 652660 + 631565 = 1284225

(d) A/R Turnover

2006 = $\frac{2300,000}{1090020}$

= 2 times.

2007 = $\frac{2,180,178}{1284225}$

= 1.6 Times

Account Receivable Turnover in days

2006 = $\frac{1090020}{2,300,000} \times 365$

= 173 days

2007 = $\frac{1284225}{2,180,178} \times 365$

= 215 days

Day's Sales in Account Receivables

2006 = $\frac{552660}{2300,000} \times 365$

= 88 days.

2007 = $\frac{631565}{2,180,178} \times 365$

= 106 days

10

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ceivable Turnover at Beginning :-

$$2006 = \frac{537360}{2300,000} \times 365$$

$= 85$ days

$$2007 = \frac{652460}{2180178} \times 365$$

$= 109.266$ days

Question 11:-

(i) Account Receivable Turnover=

$$\text{Average Inventory} = \frac{280,000 + 255,000}{2}$$

$= 267500$

$$= \frac{3050,600}{267500}$$

$= 11.4$

(ii) Account Receivable turnover in days $= \frac{267500}{3050600} \times 365$

$= 32$ days

(iii) Days Account Receivable turnover $= \frac{255,000}{3050600} \times 365$

$= 31$ days.

Q.

(iv) Inventory turnover $= \frac{2185100}{\frac{(565,000+523,000)}{2}}$

$= 4$ days

(V) Inventory turnover in days $= \frac{544,000}{2185100} \times 365$

$= 90.8$ days

(vi) Days Sales in Inventory $= \frac{523000}{2185100} \times 365$

$= 87$ days.

(vii) operating cycle $= 32 + 91$

$= 123$ days.

Azka Ehsan

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BSBA - 'C'

ASSIGNMENT: 03 :-

108/110

A

Que 1:- Accounts Recievable turnover = $\frac{\text{Net Sales}}{\text{Average A/R}}$

2002 = 17000/1700 = 10 times.

2003 = 14,580/1620 = 9 times.

2004 = 9600/1600 = 6 times.

2005 = 9000/1800 = 5 times

Account Recievable turnover in days = $\frac{\text{Average A/R}}{\text{Net Sales}} \times 365$

2002 = 1700/17000 X365 = 36.5 days.

2003 = 1620/14,580 X 365 = 40.6 days.

2004 = 1600/9600 X 365 = 60.8 days.

2005 = 1800/9000 X365 = 73 days.

?

Performance trend:-

→ Account/R turnover ratio has decreased 10 times from 2002 to 2005

→ A/R turnover in days has increased from 36.5 to 73 days.

Que 2:- Average A/R = $\frac{\text{Begining} + \text{Ending}}{2}$

$= \frac{160,000 + 180,000}{2} = 170,000$

A/R turnover in days = $\frac{\text{~~Net Sales~~ Avg A/R}}{\text{~~Average A/R~~ Net Sales}} \times 365 = \frac{170,000}{3150,000} \times 365 = 19.7$ days

A/R turnover = $\frac{\text{Net Sales}}{\text{Average A/R}} = \frac{3150,000}{170,000} = 18.53$

Day Sales in recievables = $\frac{\text{Ending recievable}}{\text{Net Sales}} \times 365 = \frac{180,000}{3150,000} \times 365 = 20.9$ days

$$\text{Average Inventory} = \frac{\text{Begining} + \text{Ending}}{2}$$

$$= \frac{390,000 + 480,000}{2} = 435,000$$

$$\text{Inventory turnover} = \frac{C.G.S}{\text{Avg Inventory}} = \frac{2250,000}{435,000} = 5.17 \text{ times.}$$

$$\text{Inventory turnover in days} = \frac{\text{Avg Inventory}}{C.G.S} \times 365$$

$$= \frac{435000}{2250,000} \times 365 = 70.6 \text{ days.}$$

$$\text{Day Sales in Inventory} = \frac{\text{Ending Inventory}}{C.G.S} \times 365$$

$$= \frac{480,000}{2250,000} \times 365 = 77.9 \text{ days.}$$

Operating cycle = A/R turnover in days + Inventory turnover in days.

$$= 19.7 + 70.6$$

$$= 90.3 \text{ days.}$$

Que 3:- Average Inventory

$$= \frac{\text{Opening} + \text{closing}}{2}$$

2

$$(1987) = \frac{(173,999 + 195,512)}{2} = 184,756.$$

$$(1986) = \frac{(167,286 + 173,999)}{2} = 170,643.$$

$$\text{Inventory turnover} = \frac{C.G.S}{\text{Avg Inventory}}.$$

$$1987 = \frac{786523}{184756} = 4.26 \text{ times}$$

$$1986 = \frac{700,263}{170,643} = 4.10 \text{ times}$$

$$\text{Inventory turnover in days} = \frac{\text{Avg Inventory}}{C.G.S} \times 365$$

$$1987 = \frac{365}{4.26} \times 365 = 8.31,273 \text{ day}$$

$$1986 = \frac{365}{4.10} \times 365 = 546 \text{ days}$$

Scanned with CamScanner

Day Sales in Inventory = $\frac{\text{Closing Inventory}}{\text{COGS}} \times 365$

1987 $= \frac{195,512}{786,573} = 90.7$

1986 $= \frac{173,999}{700,263} = 90.7$ days.

Working capital turnover = $\frac{\text{Net Sales}}{\text{Avg Working capital}}$

Average W.C $= \frac{\text{Opening} + \text{Closing}}{2}$

1987 $= \frac{241,952 + 266586}{2} = 254,269$.

$= 1086944 / 1254269 = 4.28$ times.

1986 $= \frac{217848 + 241,952}{2} = 229,900$.

$= 988,417 / 229,900 = 4.30$ times.

Working capital turnover in days = $\frac{\text{Avg. WC}}{\text{Net Sales}} \times 365$

1987 = 85.3 days.

1986 = 84.9 days.

Day Sales in Working capital = $\frac{\text{Closing W.C}}{\text{Net Sales}} \times 365$

1987 $= \frac{266,586}{1,086,944} \times 365 = 89.6$ days.

1986 $= \frac{241,952}{988,417} \times 365 = 89.3$ days.

Que 4:- COGS = Sales − G.P

= 125000 − 25000

= 100,000

Closing Stock = Opening + Purchases − COGS

= 20,000 + 110,000 − 100,000

= 30,000

$$= \frac{}{125000}$$

$= 73$ days

A/P turnover $= \frac{\text{Purchases}}{\text{Avg A/P}}$

$= \frac{110,000}{22,000} = 5$ times

A/P turnover in days $= 73$ days

Inventory turnover $= \frac{100,000}{25000} = 4$ times.

Avg. Inventory $= \frac{20,000 + 30,000}{2} = 25000$

Inventory turnover in days $= \frac{25000}{100,000} \times 365 = 91.25$ days

Financial Health:-

Company takes 73 days to collect receivables, which is quite long. 91 days of inventory on-hand indicates slow sales.

Ques 5:- Total Assets turnover $= \frac{\text{Net Sales}}{\text{Avg total Assets}}$.

Year 1 $= \frac{2050,000}{230,000} = 4.57$ times.

Year 2 $= \frac{1000\,00}{200,000} = 5$ times.

Total Assets turnover increased from 4.57 to 5 company becomes more efficient in utilizing assets.

Ques 6:- Inventory turnover measures how efficiently a company manages its inventory.

Inventory turnover $= \frac{\text{C.G.S}}{\text{Avg. Inventory}}$.

2002	2003	2004	2005
$= \frac{15000}{1000}$	$= \frac{9600}{1000}$	$= \frac{14680}{1620}$	$\frac{17000}{1740}$
$= 15$ times	$= 9.6$ times	$= 9$ times	$= 9.77$ times

Turnover $= \frac{1000}{15000}$ $= 24.3$

Interpretat.

turnover in days

$= \frac{1000}{15000} \times 365 \qquad = \frac{1000}{9600} \times 365 \qquad = \frac{1620}{14580} \times 365 \qquad = \frac{1740}{17000} \times 365$

$= 24.3$ days $\qquad = 38$ days $\qquad = 40.6$ days $\qquad = 37.4$ days

Interpretation :-

Inventory turnover decreased from 15 to 9.77, company is taking longer to sell its stock.

Inventory turnover in days increased, company is holding into company longer.

Que 7:- A/R turnover $= \frac{Sales}{Avg\ A/R}$

(2010) = 120,000 + 110,000 / 2 = 115,000.

= 1500,000 / 115000 = 13.04 times.

(2011) = 110,000 + 105000/2 = 107,500

= 1400,000 / 107,500 = 13.02 times

A/R turnover in days = $\frac{Avg\ A/R}{Sales} \times 365$

(2010) = 115000 / 1500,000 × 365 = 28 days.

(2011) = 107500 / 1400,000 × 365 = 28 days

Inventory turnover = $\frac{C.G.S}{Avg.\ Inventory}$

(2010) = 1020,000 / 240,000 = 4.25 times

(2011) = 1120,000 / 225,000 = 4.98 times.

Inventory turnover in days = $\frac{240,000}{1020,000} \times 365$

(2010) = 85.9 days.

(2011) = $\frac{225000}{1120,000} \times 365$

= 73.3 days

Operating cycle = A/R turnover in days + Inventory turn in days

(2010) = 28 + 85.9 = 113.9 days

(2011) = 28 + 73.3 = 101.3 days.

Interpretation :-

Efficiency improved in 2011

Que 08:-

C.O.G.S = Sales - Debtors
= 3100,000 - 1725000
= 1375000

Debtors turnover = $\frac{Sales}{Avg\ Debtors}$

$= \frac{3100,000}{770,000}$ = 4.03 times.

Debtors collection period = $\frac{365}{4.03}$ = 90.6 days

Inventory turnover = C.G.S / Avg. Inventory
= 13750000 / 310,000
= 4.43 times

Inventory turnover in days = $\frac{365}{4.43}$ = 82.4 days

Operating cycle = Debtors collection period + Inventory turnover in days
= 90.6 + 82.4
= 173 days

Interpretation :-

Operating cycle of 173 days
Cash is for long time.

Que 9:- Gross A/R = Recievable + Allowenes for losses

(2011) = 240,360 + 12300
352660
= 220385 + 11180
= 2315615

(2010)

= 230,180 + 7180 = 240 360 + 12300

= 237360 = 252600

Avg Gross Recievables :-

(2011) $= \frac{252660 + 231565}{2}$ = 242,113

(2010) = $\frac{237360 + 252660}{2}$ = 245,010

A/R turn over = $\frac{\text{Net Sales}}{\text{Avg. gross recivable.}}$

(2011) = $\frac{1180,178}{242 \cdot 113}$ = 4.87 tims

(2010) = $\frac{2200,000}{245,010}$ = 8.98 tims.

A/R turn over in days :-

(2011) = 365 / 4.87 = 75 days.

(2010) = 365 / 8.98 = 40.6 days.

Que 20 :- Average Gross Recievable =

2007 = $\frac{652,600 + 631,565}{2}$ = 642,113

2008 = $\frac{537,360 + 552,660}{2}$ = 545,010.

10

A/R turnover =

2007 = $\frac{2,180,178}{642,113}$ = 3.40 tims

2008 = $\frac{2300,000}{545,010}$ = 4.22 tims

A/R turn over in days =

2007 = $\frac{365}{3.40}$ = 107.4 days

2006 = $\frac{365}{4.22}$ = 86.5 days

Interpretation :-

Recievable turnover fell from 4.22 to 3.40 tims indicating slower cash flow or liquidity issues.

Que 11:- A/R turnover

$= 3050,000 / 267,500$

$= 11.11$ times

A/R turnover in days $= 267500 / 3050,000 \times 365$

$= 32$ days

Inventory turnover $= 2185,100 / 544000$

$= 4.02$ times.

Inventory turnover in days $= 544000 / 2186,100 \times 365$

$= 90.8$ days

Operating cycle =

A/R turnover in days + Inventory turnover in days

$= 32 + 90.8$

$= 122.8$ days.

Interpretation :-

corporation has fast collection period but slow inventory turnover leading to long operating cycle of 123 days.